6.3 Properties of Radicals

Name _____ Period ____ Date _____

Algebra I 6.3 (30-minute plan) · Student sheet

I can…

- use the properties of radicals to simplify expressions.
- rationalize the denominator of a radical expression.
- add, subtract, and multiply radical expressions.

Vocabulary · fill in the missing word

Product Property of Radicals — $\sqrt[n]{(ab)} =$ _____

Quotient Property of Radicals — $\sqrt[n]{\frac{a}{b}} =$ _____

Simplest form — no perfect nth-power factors, no fractions in the radicand, no radicals in a _____

Rationalizing the denominator — _____ radicals from a denominator

Conjugates — _____

Like radicals — same _____ and same radicand

Part 1 · Notes and examples (fill in each note, then work the example)

Product Property: $\sqrt[n]{(ab)} =$ _____ **Quotient Property:** $\sqrt[n]{(a/b)} =$ _____

Simplest form: A radical is in simplest form when there are no perfect nth-power _____, no _____ in the radicand, and no radical in the _____.

Example 1

Simplify. Assume all variables are positive.

a. $\sqrt{108}$ b. $\sqrt{9x^3}$

Rationalizing the denominator: To rationalize $5/\sqrt{3}$, multiply by _____.

Example 2

Simplify $\frac{5}{\sqrt{3}}$.

Conjugates: The conjugate of $\sqrt{3} + 1$ is _____.

Like radicals have the same _____ and the same _____.

Example 3

Simplify $\frac{2}{\sqrt{3}+1}$.

Part 2 · Partner practice

Partner practice

Simplify $\sqrt{72}$.